

Supplementary Materials

Article:

A Grammar-Based Criterion for Learning Sufficiency in Motion

MDPI Modeling

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Table S1. Motion-processing parameters used in the experimental setup for Scenarios E1 and E4.

Parameter	Description	Scenario 1	Scenario 4
E_{factor}	Image scalation factor to reduce resolution	1	0.5
f_v	Acquisition fps for Scenarios.	30	30
α	Decay factor used in the motion accumulation stage.	0.9	0.9
t_m	Motion timescale parameter for Scenarios $step/f_v$.	0.375	1
Pixel-level motion threshold (δ)	Threshold applied after median filtering to obtain the binary motion mask.	$35^\dagger$	$35^\dagger$
Global motion activation threshold	Threshold applied to the accumulated motion before region extraction.	$85000^\ddagger \times E_{factor}$	$85000^\ddagger \times E_{factor}$
Gaussian smoothing parameter (σ)	Standard deviation of the Gaussian filter used during frame preprocessing.	$1^\S$	$1^\S$
Gradient kernel	Operator used to estimate spatial gradients.	3×3	3×3
Median filter window	Window size used for median filtering.	5×5	5×5
Morphological structuring element	Structuring element used in the closing stage.	7×7	7×7
Minimum blob area	Minimum connected-region area retained after motion detection.	15	15

$\dagger$ Implemented as `lm_filt > 35`

$\ddagger$ Implemented as `motion_sum > 85000`.

$\S$ This value corresponds to Gaussian image smoothing in preprocessing.

Table S2. Grammar-learning parameters used in the experimental setup for Scenario E1.

Parameter	Description	Value	Usage
Minimum trajectory length	Minimum number of points required to retain a trajectory.	> 3 points	Filters short or noisy trajectories before the grammar inference stage
Track matching Threshold	Maximum admissible cost for associating detections with existing trajectories.	18	Controls trajectory continuity during frame-to-frame association
Dead-frame limit	Maximum number of frames for which a trajectory may remain unmatched before termination.	80	Determines when an inactive trajectory is removed from the active set.
Grammar extraction batch size	Number of completed trajectories accumulated before updating the grammar.	40	Defines the periodicity of grammar extraction and rule updating.
Training stopping criterion	Criterion used to stop training.	$\rho \leq 1$	Defines when the learning process is considered sufficient
Minimum number of updates before sufficiency check	Minimum number of rule-update iterations required before evaluating the stopping criterion.	> 3	Prevents early stopping before the rule set stabilizes
Rule update mechanism	Procedure used to estimate the evolving grammar	Batch update	Updates the grammar after each batch of completed trajectories.